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Palaeomagnetism of the British Carboniferous System†

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Abstract

Superficially conflicting evidence has been found concerning the direction and sense of the earth's magnetic field in carboniferous times. Various explanations of the results are considered here. We conclude that the discrepancies in direction occur because some sites underwent secondary remagnetization during the Triassic period. If this hypothesis is correct, the mean direction for the Carboniferous period was 200° east of true north and 27° down; placing Britain 15° south of the equator. As regards the sense of the field we conclude that it probably underwent a rapid series of inversions in the early part of the Carboniferous period and then remained reversed for a considerable time.

§ 1. Introduction

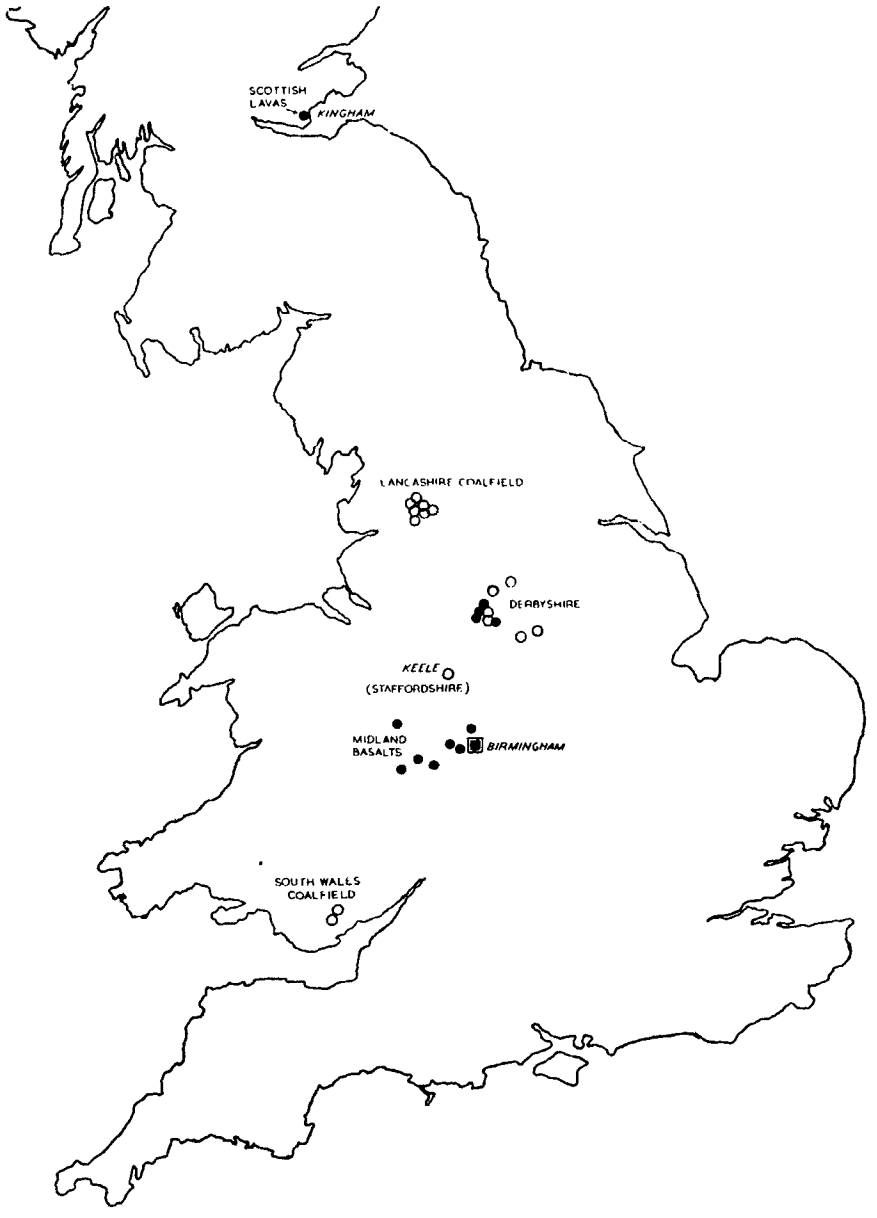
We have been independently engaged on palaeomagnetic studies of rocks from the British Carboniferous system. The investigations revealed systematic discrepancies of about 70° between the axes of magnetization in different formations, and provided strongly conflicting evidence concerning reversal of the earth's magnetic field. In the present account we propose to summarize the results and examine the discrepancies.

A map of the areas examined is given in fig. 1. They may conveniently be divided as follows: first, sedimentary formations in Derbyshire, Lancashire, Staffordshire, and South Wales; second, baked and igneous formations in the English Midlands including Derbyshire; third, baked and igneous formations in Scotland.

The sedimentary formations cover almost the complete Carboniferous period. In Derbyshire, the geological succession comprised about 14 000 ft of sediments extending from the Carboniferous limestone to the Coal Measures; the estimated time-span being about 220–280 million years. In the Lancashire Coalfield there are 20 000 ft of sediments, and in the South Wales Coalfield 15 000 ft of sediments, covering approximately the same period. Samples were taken from 46 sites in these areas, and from one site in the late Carboniferous Keele beds of Staffordshire. Further details are given in the table.

† Communicated by the Authors.

Fig. 1



Distribution of sites.
● Baked and igneous formations.
○ Sedimentary formations.

Summary of Results

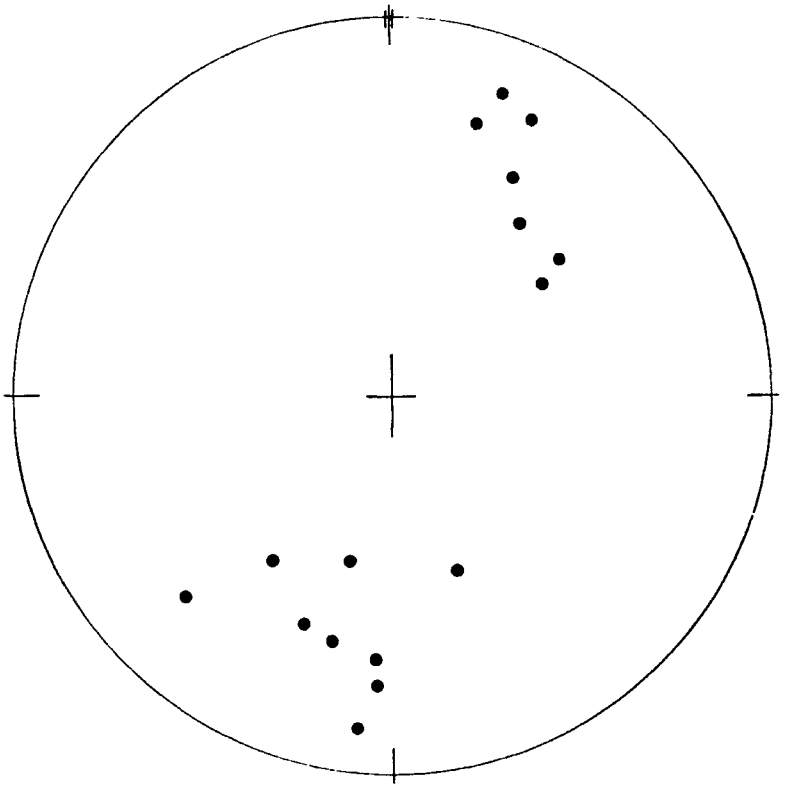
Region	Description	Material examined		Material accepted		Declination	Inclination	Period	Approximate age
		Sites	Samples	Sites	Samples	Degrees east of true north			Millions of years
Derbyshire	Sedimentary	22	84	3	13	37°	39° down	Carboniferous Limestone to Millstone Grit	230–280
				3	11	197°	37° down		
Lancashire	Sedimentary	12	49	4	14	24°	22° down		
				3	10	184°	23° down		
Staffordshire	Sedimentary	1	7	1	7	199°	34° down	Late Carboniferous	270
South Wales	Sedimentary	12	79	2	18	205°	49° down		
English Midlands	Baked Sills	6	11	6	11	202°	17° down	Late Carboniferous	280
			12		12	196°	13° down		
	Baked Lava	1	2	1	2	349°	1° down	Carboniferous Limestone	240
			5		3	357°	14° down		
Derbyshire	Baked Sill (1)	1	6	1	6	221°	34° down	Middle Carboniferous?	260?
			7		7	218°	36° down		
	Lava (2)	1	3	1	3	48°	45° up	Carboniferous Limestone	240
	Lavas (2)	2	3	2	3	200°	29° down		
(2) (3) Scotland (Kinghorn)	Beds 9–15	4	7	2	5	12°	18° up	Calciferous Sandstone	230
	Beds 16–20	4	11	3	8	170°	24° down		
	Bed 21	1	3	1	2	55°	19° down		
	Beds 25–48	13	25	10	20	13°	35° up		
	Beds 50–55	5	8	4	6	205°	41° down		
	Bed 65	1	2	1	2	15°	14° down		
	Bed 69 (4)	1	2	1	2	15°	35° up	Late Calciferous Sandstone?	240?

(1) The origin of this formation at Tideswelldale is not certain. It may be lava.
(2) The directions for two of the Derbyshire lavas and in beds 12, 15, 19, 20, 31, 32, 38, 48 from Kinghorn are values after thermal or alternating field cleaning (see text).
(3) All the results are from igneous rocks except beds 25, 45, 54 which were baked sandstones.
(4) This formation is a sill of uncertain age.

The English baked and igneous rocks were from two areas. In the first, near Birmingham, there were six late Carboniferous sills and one lava of Carboniferous Limestone age. In the second area, in Derbyshire, there were three lavas also of Carboniferous Limestone age, and one Middle or Late Carboniferous Sill.

The Scottish baked and igneous rocks were a series of lavas and interbedded sediments from Kinghorn, Fife, covering a few million years in the Calciferous Sandstone period. In the table the beds are numbered consecutively from the bottom upwards, irrespective of their nature, in accordance with the notation of Geikie (1900).

Fig. 2



Results from sedimentary formations (lower hemisphere of Schmidt projection).

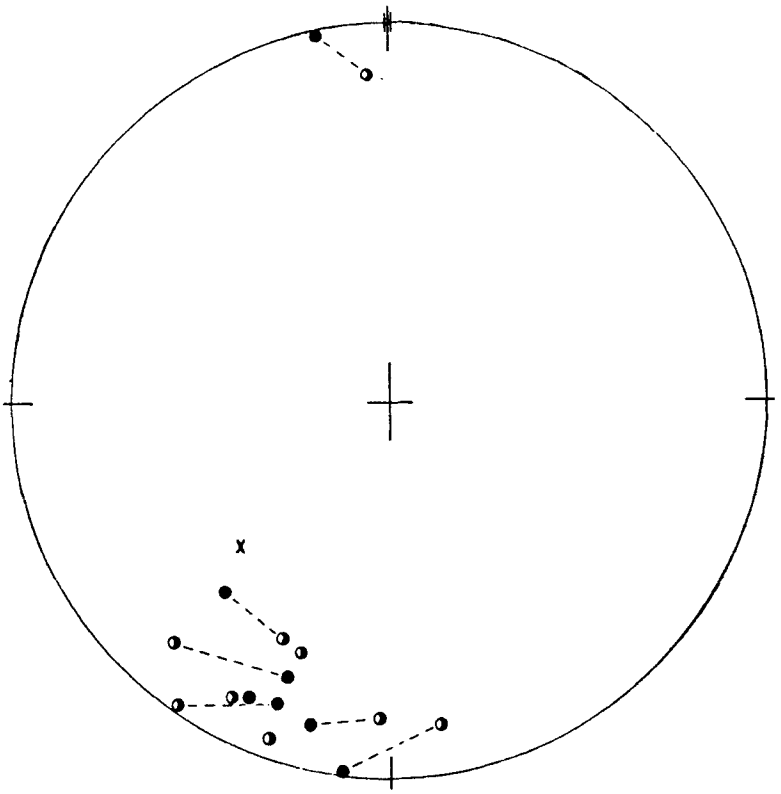
Some details of the sampling are included in the table. Several oriented blocks were collected at each site. As a rule, three to six discs $1\frac{3}{8}$ in. diameter $\frac{1}{4}$ in.–$\frac{1}{2}$ in. thick were cut from each block and measured on a Blackett magnetometer. The mean directions and, where necessary, the standard deviations were calculated by the methods given by Wilson (1959). Certain other measurements were also carried out as described in the following sections.

§ 2. Results

2.1. *Directions of Magnetization*

The main results are set out in the table and plotted on figs. 2–4. The directions are all given relative to the original bedding and flow-planes. They have been plotted on the lower hemisphere of a Schmidt projection, the sense of the magnetization being reversed for the upward dips.

Fig. 3



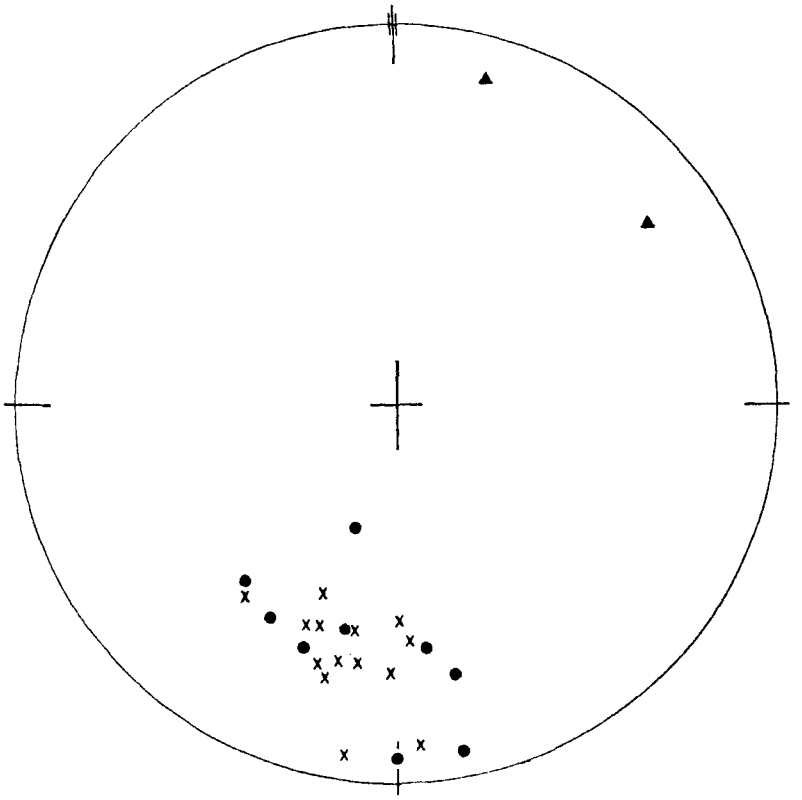
Results from English baked and igneous formations (lower hemisphere of Schmidt projection).

● Baked rocks.
◑ Igneous rocks (north magnetic pole).
× Igneous rocks (south magnetic pole).
– – – Joins results from same site.

In the sedimentary rocks the magnetization was often extremely weak and sometimes widely scattered ; but at some sites there was good agreement between the mean directions of different blocks. As a first step in analysis, only the sites for which the standard deviations between individual blocks was less than 20° were accepted. Out of 47 sites, 16 fall within this category. Their directions, which are plotted in fig. 2,

fall in two well-defined groups ; one with a mean azimuth 34° east of true north and a mean dip of 31° down, and the other 195° east and 35° down. The smaller angle between these two mean axes is about 70°. On the sites with scattered directions, the results still showed some tendency to group about the same two axes. Sometimes the two directions even occurred at the same site ; the most striking example being the Union Strip Mine, Derbyshire, where three blocks agreed closely at 25° east, 33° down and three others agreed at 213° east and 31° down.

Fig. 4



Results from Scottish baked and igneous formations (lower hemisphere of Schmidt projection).

● North pole.
× South pole.
△ Discordant directions (north pole).

Figure 3 shows the directions in the English baked and igneous rocks. In the sills and associated baked sediments the mean direction was 202° east and 19° down ; while of the lavas from Derbyshire, one was magnetized 46° east, 45° up, and two showed initial north-east downwards directions, which changed on heating to within 100°c of the Curie

temperature and gave a final mean direction of 200° east and 20° down. The remaining lava, near Birmingham, gave a mean 357° east and 7° down ; but since the scatter of directions was large (standard deviation 14°), this probably only represented a reversal of the south-west downward axis.

The directions for the baked and igneous formations in Scotland contrast with the preceding data. They are given in fig. 4. The mean axis was 193° east 33° down. Along this axis, magnetizations existed in either sense ; there being successive groups of normal and reversed lavas. In two lavas (numbers 25 and 65) lying between normal and reversed groups there were discordant directions which are differently designated in fig. 4. Apart from these, which may represent an intermediate direction during the course of an earth's field reversal, there was no sign of the north-east/downwards magnetization.

The results as a whole, then, show two distinct directions ; 40 sites being magnetized along south-east/downwards axes, and 7 along a north-east/downwards axis. The angle between these two axes is about 70°. It should be remarked that some measurements by Clegg *et al.* (1954) on sediments at Frampton Cotterill Gloucestershire, also gave a north-east/downwards direction (44° east, 35° down).

2.2. *Stability*

In assessing the significance of the magnetizations we must consider not only their stability (that is whether they represent a unique direction of the earth's magnetic field) but also their age : for a stable magnetization may easily have been acquired long after the rocks were formed.

The stability of the south-west/downwards direction is demonstrated by four lines of evidence, as follows :

(1) There was close agreement in direction between igneous and sedimentary rocks of widely different type.

(2) Exact 180° reversals occurred ; showing that there had not been any secondary magnetization sufficient to disalign the two directions.

(3) The directions for the Scottish beds (which are tilted 30° eastwards) agreed with the English ones more closely relative to the bedding planes than the present horizontal. This indicates that the magnetization was acquired before tilting.

(4) There was good agreement between the directions in associated baked and igneous rocks. This line of evidence seems especially convincing since the intrinsic scatter of the baked rocks was only 2–3°.

The measurements on baked rocks also provided a test of the date of magnetization. At several sites there was a large and systematic decrease of intensity and increase of scatter with distance from the contact (Everitt and Clegg 1960), which strongly suggests that the baked material became magnetized when the igneous bodies cooled, and therefore that the magnetizations were of Carboniferous age.

Several of the Scottish lavas had widely scattered natural magnetizations. Laboratory heating experiments were carried out on six of these by Mr. R. L. Wilson. In each case, he found that the directions came into line with the mean values for adjacent beds, on heating to within 100°C of the Curie temperature. For two lavas a similar effect was achieved by alternating field demagnetization. The sites treated in this way are distinguished in a note to the table.

The stability of the north-east/downwards magnetization is not so well attested. However, there is some field evidence from measurements on two limbs of the Pendle monocline, which agreed more closely relative to the bedding-planes than to the present horizontal ; although since the direction of magnetization was only a few degrees from the strike axis, this test is not absolutely decisive. In addition, there was some laboratory evidence for stability, since the directions of magnetization showed no appreciable change on heating in zero field to 200°C.

2.3. *Age of the Magnetizations*

The evidence from baked rocks, outlined in the preceding section, strongly suggests that the south-west/downwards magnetizations were acquired at the time of formation of the igneous rocks, which—for the Kinghorn lavas at least—can confidently be assigned to the Carboniferous period.

The north-west/downwards magnetization on the other hand occurs only in the sedimentary rocks and may be assigned to any period before the folding of the Pendle monocline. The mean direction was 39° east and 35° down, which is remarkably near the figure of 34° east, 28° down given by Clegg *et al.* (1954) for the Triassic period. It is therefore conceivable that the magnetization was acquired by a secondary process during Triassic times.

There is some geological evidence to support this view. Many of the Carboniferous formations underlying the Permo-Triassic strata on either side of the Pennines are markedly reddened to depths of 1000 ft or so. According to Trotter (1953), the reddening was due to weathering of the iron minerals to hematite while the overlying strata were being formed. The beds examined here did not belong to the most severely weathered series but, as we have seen, the magnetic material in them was hematite. Although it has not been possible to relate in detail their position to the Permo-Triassic erosion surface, there does seem to be a definite connection. This may be illustrated conveniently by results from three sites in the Coal Measures of Derbyshire–Yorkshire border. At Sheffield Brick Pit, which is some way from the erosion surface, the mean direction was 207° east, 36° down ; at Hardwick Hall, which is very near the erosion surface, the direction was 50° west and 33° down ; while at Union Strip Mine, which was slightly further away than Hardwick Hall, three blocks agreed at 25° east, 33° down, and three others agreed at 213° east,

31° down. Much the simplest explanation of these results is in terms of local re-magnetization during Triassic times.

A second line of evidence is provided by the Keele beds of Staffordshire, in which the hematite is thought to be primary and should therefore retain the original Carboniferous magnetization. Their direction was indeed 199° east and 34° down.

The most obvious objection to the interpretation in terms of remagnetization is that the beds of the Pendle monocline appear to have acquired their north-east magnetization before being folded in the Hercynian orogeny, which occurred during early Permo-Triassic times. As mentioned above, the evidence is not absolutely decisive. But even if the beds were magnetized before folding, that only proves that the north-east downwards direction appeared fairly early in Permo-Triassic times : it may still have been a considerable time after the Carboniferous period ended.

2.4. *Reversals*

We have seen that at Kinghorn the normal and reversed magnetizations occur in separate groups of five or more lavas. This suggests that the earth's field underwent inversions : a view which is supported by the conjunction of concordantly magnetized baked sediments with both normal and reversed groups : and also by the absence of any positive evidence from the heating experiments for physico-chemical reversals. If so, inversions of the field occurred at the rate of about one per million years—that is, at about the same rate as found by Hospers (1953) for the Tertiary period.

If this interpretation is correct, it is puzzling that all the lavas and baked sediments showing these periodic inversions were formed during a relatively small part of the early Carboniferous, while all the later sediments and intrusions (which extend over the major part of the period) are reversed.

In assessing these results, we distinguished two questions. The first is whether the earth's field reversed at all, the second whether it underwent a rapid sequence of inversions from normal to reversed and back. There seem to be four possibilities :

(i) All the sedimentary and some igneous rocks were self-reversed.

(ii) All rocks were magnetized in a reversed field, and some early lavas and associated baked sediments underwent physico-chemical reversal to the normal direction.

(iii) Inversions of the earth's field occurred frequently throughout the period, but the sediments and intrusions by chance only became magnetized during periods of reversed field.

(iv) Inversions occurred frequently during early Carboniferous times, but afterwards the field remained continuously reversed for a considerable period.

The first explanation would require a great exercise of special pleading. Of the remaining explanations the second is the least satisfying because once the possibility of a reversed field is admitted, the main ground for believing that self-obliterating physico-chemical reversals have occurred commonly in rocks disappears. The third hypothesis is excluded, because it seems inconceivable that all the sediments became magnetized in less than a million years. Hence it is best to accept tentatively the fourth hypothesis that the earth's field underwent several inversions during the early part of the period and afterwards remained reversed.

§ 3. Conclusion

Our conclusions may be summarized as follows :

1. The results in the south-west/downwards group give a mean axis of magnetization for the Carboniferous period (age about 250 million years) of 200° east and 27° down. The standard deviation was 20°. These figures are consistent with the directions of 200° east and 35° down given by Nairn (private communication) for the continent of Europe, and 189° east and 9° up given by Creer (1957) for the early Permian in Britain (age about 200 million years).

Some of our results have already been quoted by Irving (1959) and by Blackett, Clegg and Stubbs (1960). The present results are based on more complete and accurate data, and must be regarded as superseding their figures. Broadly they confirm the picture presented by these workers but place Britain about 20° further south, and thus, for example, considerably reduce the relative displacement between England and Australia from that originally claimed by Irving.

Our final mean values for the four quantities ΔI, ϕ, ψ and λ_m defined by Blackett *et al.* (1960) are ΔI, 92° ; ϕ, 87° ; ψ, 23° ; and λ_m, 22° S.

2. Some of the sediments were probably remagnetized during Triassic times. This suggests that the effects of Permo-Triassic weathering extended to a considerably greater depth than has usually been supposed. Palaeomagnetic measurements obviously provide a powerful method of investigating such phenomena.

3. The earth's field seems to have undergone a rapid series of inversions in early Carboniferous times and afterwards to have remained reversed for a considerable period.

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